GABBER'S THEOREM: FOR A HENSELIAN PAIR THE ETALE COHOMOLOGY OF A TORSION SHEAF ON SPEC A AGREES WITH THAT ON SPEC A/I

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the induction on the degree, the comparison of the two long exact sequences, and the two effacement arguments (with Z = X for injectivity and with Z = Z for surjectivity) are all present, with the degree q-1 case applied in each half.

Editorial note. The label theorem-gabber names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own theorem environment in the chapter source.

Editorial note. The statement is an author Theorem environment (named Gabber in the source), delivered verbatim with its label; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Henselian pair (A, I) and the closed immersion Z = Spec(A/I) into X = Spec(A). Same chapter, the notion of a henselian pair and the h^0 statement for the pair of schemes Z subset X. Supplies the geometric set-up and the vanishing lemma giving the degree zero case.
- Torsion abelian sheaves on X_etale and restriction along Z -> X. Same chapter, the category of torsion abelian sheaves on the etale site and the functor F |-> F|_Z. Supplies the coefficient sheaves F, F', Q and the restricted sheaves appearing in the two long exact sequences.
- Effacement of a cohomology class by passing to a larger torsion sheaf through a finite cover. Same chapter, the lemma effacing the cohomology of a closed subscheme by a finite cover. Supplies the choice of the injection F -> F' making an arbitrary class die in the next term, in both halves of the induction step.

Source reference. Original reference 09ZE: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14901–14901. Prerequisite pointer invoked at proof

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line(s) 15241; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09ZH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15180–15180. Prerequisite pointer invoked at proof line(s) 15212; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Gabber)**.** *Let (A, I) be a henselian pair. Set $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(A/I)$. For any torsion abelian sheaf $\mathcal{F}$ on $X_{\acute{e}tale}$ we have $H^q_{\acute{e}tale}(X, \mathcal{F}) = H^q_{\acute{e}tale}(Z, \mathcal{F}|_Z)$.*

Proof. The result holds for $q = 0$ by Lemma 09ZH. Let $q \geq 1$. Suppose the result has been shown in all degrees $< q$. Let $\mathcal{F}$ be a torsion abelian sheaf. Let $\mathcal{F} \to \mathcal{F}'$ be an injective map of torsion abelian sheaves (to be chosen later) with cokernel $\mathcal{Q}$ so that we have the short exact sequence

$$0 \to \mathcal{F} \to \mathcal{F}' \to \mathcal{Q} \to 0$$

of torsion abelian sheaves on $X_{\acute{e}tale}$. This gives a map of long exact cohomology sequences over X and Z part of which looks like

$$\begin{array}{ccccccc}
H^{q-1}_{\acute{e}tale}(X, \mathcal{F}') & \longrightarrow & H^{q-1}_{\acute{e}tale}(X, \mathcal{Q}) & \longrightarrow & H^q_{\acute{e}tale}(X, \mathcal{F}) & \longrightarrow & H^q_{\acute{e}tale}(X, \mathcal{F}') \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
H^{q-1}_{\acute{e}tale}(Z, \mathcal{F}'|_Z) & \longrightarrow & H^{q-1}_{\acute{e}tale}(Z, \mathcal{Q}|_Z) & \longrightarrow & H^q_{\acute{e}tale}(Z, \mathcal{F}|_Z) & \longrightarrow & H^q_{\acute{e}tale}(Z, \mathcal{F}'|_Z)
\end{array}$$

Using this commutative diagram of abelian groups with exact rows we will finish the proof.

Injectivity for $\mathcal{F}$. Let ξ be a nonzero element of $H^q_{\acute{e}tale}(X, \mathcal{F})$. By Lemma 09ZE applied with $Z = X$ (!) we can find $\mathcal{F} \subset \mathcal{F}'$ such that ξ maps to zero to the right. Then ξ is the image of an element of $H^{q-1}_{\acute{e}tale}(X, \mathcal{Q})$ and bijectivity for $q - 1$ implies ξ does not map to zero in $H^q_{\acute{e}tale}(Z, \mathcal{F}|_Z)$.

Surjectivity for $\mathcal{F}$. Let ξ be an element of $H^q_{\acute{e}tale}(Z, \mathcal{F}|_Z)$. By Lemma 09ZE applied with $Z = Z$ we can find $\mathcal{F} \subset \mathcal{F}'$ such that ξ maps to zero to the right. Then ξ is the image of an element of $H^{q-1}_{\acute{e}tale}(Z, \mathcal{Q}|_Z)$ and bijectivity for $q - 1$ implies ξ is in the image of the vertical map. □